

Berkeley and Heiserman as an Unexhausted Architecture for Embodied Machine Intelligence

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Abstract

Edmund C. Berkeley is usually remembered as a writer who helped connect symbolic logic to computing machinery. That description is correct, but incomplete. Read across Berkeley’s machine-oriented writings and projects, the central concern is broader: to show that symbolic logic can serve as a practical design language for machines that acquire information, retain it, respond appropriately to changing conditions, and organize their behavior over time. This paper takes *Symbolic Logic and Intelligent Machines* as the principal text in that effort, while treating it as representative of a larger Berkeley program that links logical form, circuitry, control, and intelligent behavior. On this reading, Berkeley does not merely treat intelligence as abstract symbol manipulation. He repeatedly defines intelligent machines in operational terms, ties logic to hardware realization, and describes machine behavior through the coordinated interaction of inputs, outputs, memory, calculation, control, states, and events. His treatment of robots and machine activities accordingly moves beyond static logical form toward temporally extended, environment-coupled behavior. David L. Heiserman’s machine-intelligence work extends this program toward adaptive creature architectures built around memory, confidence, and generalization. Taken together, Berkeley and Heiserman can be read not as exhausted historical episodes but as contributors to a still under-tested architectural approach to embodied robotic cognition.

1 Introduction

Contemporary AI and robotics research often oscillates between large-scale statistical learning, abstract planning systems, and highly specialized control solutions. In that environment, behavior-organizing approaches rooted in explicit states, logical conditions, hardware-near control, and modular sensorimotor structure can be dismissed too quickly as dated. This paper begins from the opposite suspicion: that these design commitments remain fresh, technically coherent, and experimentally available for robotic cognition.

Edmund C. Berkeley is commonly placed in the history of symbolic logic, early computing, and machine thought. That placement is justified. *Symbolic Logic and Intelligent Machines* is overtly concerned with Boolean algebra, logical relations, class reasoning, and the design of information-handling machinery [1]. Yet this standard placement risks narrowing Berkeley’s significance. He was not only interested in the formal calculability of statements and classes. He was also interested in how reason-like competence could be built into machinery that senses, stores, decides, and acts.

The aim of this paper is to argue that Berkeley should be read as an architect of embodied machine intelligence rather than only as a symbolic logician of machines. He is not a full theorist of adaptive embodied AI in the Heiserman sense, but neither is he adequately understood as a

purely formal or disembodied thinker. His work gives a logically rigorous account of how physically organized machines can take in information, retain it, operate on it, and guide their own activity over time [1, pp. 66–70]. When read alongside Heiserman’s creature-based machine intelligence, Berkeley becomes part of a still-open design program rather than a closed episode in the history of ideas.

2 Research Question and Thesis

The guiding question is how Berkeley’s *Symbolic Logic and Intelligent Machines* should be read today: merely as a period work of symbolic logic applied to machinery, or as a still-usable account of embodied, behavior-organizing intelligence.

The working thesis is that Berkeley’s work is best read as a technically viable account of embodied machine intelligence whose implications have not been exhausted. He inherits formal machinery from Boole, Venn, and Shannon, yet uses that inheritance to theorize intelligent systems as physically realized, temporally organized, and behaviorally active machines [2–5]. Heiserman extends the same broad design space toward adaptive creature intelligence organized around memory, confidence, and generalization [6]. More strongly, Berkeley and Heiserman together can be read as defining an underused architectural alternative whose engineering potential has not been fully tested against current assumptions about how robotic intelligence must be built. The literature review below is organized to support that stronger claim by showing that Berkeley is best understood neither as a routine symbolic logician nor as an isolated precursor, but as a figure whose work becomes sharper when placed among adjacent traditions in formal logic, cybernetics, adaptive machinery, and embodied control.

3 Literature Review

The literature relevant to this paper falls into five overlapping bodies of literature. The first and most important consists of the primary sources that directly carry the argument: Berkeley, Boole, Venn, Shannon, and Heiserman. Berkeley is the principal object of interpretation, but the point of the paper is not recoverable from Berkeley alone. Boole provides the underlying claim that logical operations can be written in a compact symbolic calculus and manipulated with rule-governed precision [2, 3]. Berkeley does not invent his symbolic resources from scratch; he inherits a tradition in which reasoning is already being recast as explicit operation. Boole gives him the confidence that classes, statements, and operations can be made exact enough to serve as components in a design language. Venn matters in two distinct ways. In *Symbolic Logic*, he sharpens the class-logic tradition Berkeley works within and clarifies the status of logical symbolism as a general calculational language rather than a mere classroom mnemonic [4]. Berkeley’s use of symbolic logic as a practical language for machine organization sits more comfortably once that broader Vennian conception is in view. In *The Logic of Chance* and *The Principles of Empirical or Inductive Logic*,

meanwhile, Venn moves toward probability, induction, and knowledge under conditions that are not exhausted by strict deduction [7, 8]. That second Vennian strand helps distinguish Berkeley’s mostly rule-bound machine logic from Heiserman’s move toward machines that store experience, accumulate success conditions, and generalize from them. Shannon supplies the crucial bridge from logical relations to relay and switching hardware: once logical form is realizable as circuitry, symbolic structure becomes directly architectural rather than merely notational [5]. If Boole and Venn explain Berkeley’s formal inheritance, Shannon explains why Berkeley could treat logic as a machine-building discipline rather than only as a notational one. Heiserman, by contrast, is not part of Berkeley’s own bibliographic world, but functions as a comparison point that clarifies how the same broad design space can be pushed from explicit control organization toward adaptive and experiential machine intelligence [6]. His Alpha, Beta, and Gamma levels supply a vocabulary for comparing fixed behavioral organization, learned response, and confidence-bearing generalization without abandoning embodiment. Heiserman therefore does not merely extend Berkeley chronologically; he pressure-tests Berkeley conceptually by asking what happens when a state-organized creature must also remember, prefer, and generalize.

The second body of literature consists of immediate contextual works in symbolic logic, especially Basson and O’Connor’s *Introduction to Symbolic Logic* [9]. This source is worth more than a passing mention because it gives a fairly orthodox mid-century picture of what symbolic logic education looked like when centered on the propositional calculus, the algebra of classes, quantificational form, and formal deduction as such. Read against Basson and O’Connor, Berkeley’s difference becomes sharper. He is not simply teaching symbolic manipulation, validity, or proof technique. He is repeatedly trying to move symbolic logic outward into circuit design, temporal organization, state transition, memory, and machine behavior. Without a foil like Basson and O’Connor, Berkeley’s practical and architectural language can be mistaken for a routine extension of symbolic-logic pedagogy. With it, we can see that Berkeley is selecting from the symbolic tradition and reorienting it toward implementation, control, and intelligent machinery. Basson and O’Connor therefore help establish that Berkeley is unusual not because he knows different logical fundamentals, but because he asks those fundamentals to do a different kind of work.

The third body of literature is the cybernetic and control literature: Wiener, Ashby, Turing, and Walter [10–15]. Each author contributes something different. Wiener contributes the most general postwar frame of communication, control, feedback, and the parallel treatment of animal and machine systems. He helps define the field in which organized signaling and regulation become central engineering questions. Ashby’s two books are particularly important for this paper’s vocabulary. *Design for a Brain* is directly relevant because it treats adaptation, equilibrium-seeking, and self-organizing behavior as machine problems rather than as mysteries outside mechanism [11]. *An Introduction to Cybernetics* adds a more explicit language of state, transition, regulation, variety, and machine description [12], making it especially useful for comparison with Berkeley’s own state-and-event formalism. Ashby is especially important because he and Berkeley share an interest in state-organized systems while differing over emphasis: Ashby tends to abstract toward

general regulatory principles, whereas Berkeley keeps pulling the discussion back toward logical specification, circuit organization, and the engineered composition of intelligent machines. Turing enters this literature review for a different reason. In *Intelligent Machinery* and “Computing Machinery and Intelligence,” he keeps open the problem of machine intelligence as something that may require organization, education, learning, and developmental method rather than mere fixed calculation [13, 14]. His language of discrete machinery, memory capacity, and even unorganized machines gives us a second comparison class in which machine intelligence is treated as buildable, but not primarily through Berkeley’s route of explicit symbolic architecture. Turing is therefore close enough to Berkeley to be illuminating and different enough to be clarifying: Berkeley is more architectonic and control-explicit, Turing more open to training, programming, and developmental organization. Walter’s role is more concrete and embodied. His work on brain mechanisms and autonomous tortoise-like devices shows how relatively simple circuitry can produce environment-facing, apparently purposive behavior [15]. Walter matters because he demonstrates in physical form what Berkeley often argues in conceptual and architectural terms: that intelligence can appear at the level of organized sensorimotor behavior without requiring disembodied rationality. Taken together, these works do not prove Berkeley’s argument, but they do show that Berkeley belongs in a live postwar conversation about regulation, autonomy, adaptation, and machine behavior, not only in the narrower history of formal logic.

The fourth body of literature is behavior-based or embodied-machine comparison, especially Braitenberg and Brooks [16–18]. These references are used cautiously, but they should not be treated lightly. Braitenberg is valuable because he shows how surprisingly rich behavior can arise from very simple sensorimotor couplings. This keeps the paper from assuming that intelligence must begin at the level of high symbolic representation. Brooks matters because he explicitly argues for situated, layered, behavior-producing architectures that do not depend on heavy centralized representation. The point is not that Berkeley anticipates Brooks in a straightforward way; he does not. Berkeley is much more willing than either Braitenberg or Brooks to preserve an explicit language of logical condition, internal organization, and designed control. Yet the comparison is still productive because it reveals that Berkeley’s machines are not static theorem engines. They are systems that sense, transition, remember, and act in time. The later behavior-based literature therefore does two things at once: it prevents us from mistaking Berkeley for a contemporary behavior-based roboticist, and it prevents us from misclassifying him as a purely disembodied symbolist. The resulting picture is more exact. Berkeley occupies a middle position in which symbolic organization and embodied control are not adversaries but cooperating layers of machine intelligence.

The fifth and lightest body of literature is secondary historical or philosophical interpretation. This paper uses such literature only sparingly. Its method is primarily analytic and primary-source driven. The goal is not to subsume Berkeley under an already settled historiography, but to show from the texts themselves that his work occupies a more behavior-centered and embodied position than is usually granted.

This review also clarifies the paper’s novelty claim. We have not identified a prior literature that explicitly joins Berkeley and Heiserman in a single embodied-machine-intelligence architecture. The existing literature gives us Berkeley as a figure in symbolic logic and machine thought, Heiserman as an experimenter in adaptive robot intelligence, and recent work on state-based robotic behavior as a separate contemporary concern. What appears to be missing is a publication that explicitly argues that Berkeley and Heiserman belong to the same technically significant line of development. The point of assembling these five bodies of literature is therefore not just to provide background. It is to show why that linkage is interpretable, technically serious, and worth stating as a live architectural claim.

From Published Work to Live Engineering Hypothesis

The argument of this paper is not that Berkeley and Heiserman merely deserve a more generous place in intellectual memory. The stronger claim is that they define a partially neglected engineering line. The literature reviewed above makes that claim more precise. In Berkeley’s case, this line joins logical structure to hardware, control, state organization, and robot behavior. In Heiserman’s case, it extends into machine creatures that accumulate experience, retain successful responses, and later generalize from them under confidence-sensitive conditions [6, pp. 18–20]. What links them is not a shared vocabulary alone, but a shared architectural intuition: intelligence is something built out of organized relations among sensing, internal state, control, memory, and outward action.

What makes this line interesting now is not antiquarian recovery but architectural coherence. Both writers assume that intelligent machinery must be organized around explicit states, condition-sensitive transitions, sensorimotor coupling, and the structured coordination of internal and outward behavior. Even where the two differ, they differ along a productive axis rather than across a conceptual break. Berkeley provides more formal and engineered control organization, while Heiserman pushes further toward adaptive behavior and memory-driven response. Read together, they make visible a design space that contemporary AI often fragments into separate topics such as symbolic reasoning, control, behavior, and learning.

This does not yet prove that their architecture offers the best path forward for modern robotics. It does, however, justify treating it as a live engineering hypothesis rather than a completed or obsolete episode. The real question is not whether their systems can simply be reproduced, but whether behavior-organizing, state-based, hardware-near architectures remain powerful in domains where deterministic timing, embodied interaction, modular design, and constrained compute still matter. If they do, then Berkeley and Heiserman are not just historically interesting. They become technically provocative again.

Logic Embodied in Machinery

One of Berkeley’s major contributions is to insist that symbolic logic is not merely a way of describing reasoning in abstraction. It is also a way of designing machinery. This is the decisive move from logic as formal analysis to logic as implemented intelligence. The bridge here is Shannon.

Once Boolean relations can be realized in relay and switching circuits, conditions and inferences no longer remain on paper or in thought alone. They become structures of hardware [5]. Berkeley's importance lies in taking that bridge seriously as an architectural principle rather than as an isolated technical trick.

Berkeley adopts this possibility as fundamental. In his treatment, classes, statements, and logical conditions become design elements for circuits, control systems, and automatic machines. Logic is therefore not simply explanatory; it is operational. It specifies distinctions a machine can embody, conditions it can test, and sequences it can regulate. Berkeley's interest in symbolic logic is inseparable from his interest in machinery capable of carrying out organized, reason-like operations in the world [1]. This is why his work reads less like a mere extension of formal logic and more like an attempt to formulate a language for intelligent-system design.

Intelligent Machines as Organized Systems

The strongest evidence for an embodied reading appears in Berkeley's own definitions and examples. He lists a wide range of machines as "intelligent machines," including traffic-light controllers, air-traffic systems, factory control equipment, game-playing machines, reading and recognizing devices, robots, and digital computers [1, pp. 66–69]. This list matters because it reveals the breadth of Berkeley's target. Intelligent machinery is not confined to formal problem solving. It includes systems coupled to signals, environments, movement, and regulated action.

Berkeley then proposes a general architecture shared by almost every intelligent machine. Such a machine, he says, has inputs, outputs, storage or memory, calculating capacity, and a control unit [1, p. 70]. Conceptually, this is one of the most important passages in the book. It frames intelligence not as a single faculty but as the coordination of sensing, acting, remembering, processing, and directing. Especially important is Berkeley's remark that the control unit may itself be subject to instructions calculated by the machine, allowing the machine in some sense to guide itself [1, p. 70]. The autonomy is limited, but it is real: control is not merely imposed from outside, but can be organized internally as part of the machine's own operation.

States, Events, and Temporally Extended Behavior

If Berkeley had stopped with Boolean algebra and machine architecture, the case for an embodied reading would remain suggestive. It becomes much stronger in his treatment of the algebra of states and events. There Berkeley explicitly argues that Boolean calculus must be extended in order to come "into closer relation with the real world," since classes and statements change with time and since machine operation involves changing states and events rather than static truths alone [1, pp. 145–147]. This is one of the clearest points at which Berkeley moves beyond a static conception of logical form.

A state lasts for a time; an event is brief. This distinction allows Berkeley to formalize persistence, interruption, transition, delay, and sequence. He also insists on the practical importance of time units, step-functions, and even intervals in which a machine state is temporarily undefined

during transition [1, pp. 145–147]. The importance of this move is difficult to overstate. Once machine activity is represented as passage through states and events, intelligent behavior becomes describable as organized temporal process rather than as a mere arrangement of timeless logical forms. In this respect Berkeley pushes logic toward an account of regulated activity, which is precisely the direction an embodied theory of intelligent machinery requires.

Robots and Behavioral Programs

Berkeley’s robot chapters make the behavioral implications even clearer. He defines a robot as a machine able to “behave” by itself, taking in sensations by means of sensing devices, performing actions by means of acting devices, and correlating sensations and actions by means of circuits or mechanisms expressing instructions [1, pp. 177–178]. That definition is already much closer to an agent-based conception of intelligence than to a purely symbolic one. The robot is not presented as a proof engine with motors attached. It is presented as a behaving system whose intelligence lies in the organization of sensation, action, and instruction.

His examples are equally revealing. Berkeley’s electronic robot squirrel “Squee” is organized around activities such as hunting, homing, and depositing, each of which persists until relevant conditions terminate it [1, pp. 177–180]. Likewise, the “robot animal” problem is introduced explicitly as a programming problem in which one must assign a number of states to a robot and then specify changes from one state to another when certain events happen [1, pp. 177, 180–181]. These are not merely symbolic demonstrations. They are behavior-organizing schemes in which logical conditions select, maintain, and redirect activity. In other words, Berkeley’s robot programming already treats intelligence as structured conduct in time.

Berkeley and Heiserman

The comparison with David Heiserman clarifies both Berkeley’s strengths and his limits. Heiserman’s *Robot Intelligence ... with Experiments* presents an evolutionary hierarchy in which Alpha-class creatures rely on constrained reflexive activity, Beta-class creatures add a memory of past successful responses, and Gamma-class creatures generalize from prior experience while building confidence in successful patterns of action [6, pp. 18–20]. In that framework, machine intelligence is lived by a creature in an environment. It senses, acts, remembers, adapts, and in the Gamma case anticipates. Heiserman thus supplies what Berkeley mostly leaves implicit: a developmental account of how embodied organization can become progressively more adaptive.

Berkeley does not reach this level of adaptivity. His machines are mostly engineered top-down rather than self-modifying through reinforcement or confidence-based generalization. He does not provide Beta-style stored response schemes or Gamma-style conjectural extensions of prior success. Even so, the continuity remains important. Berkeley already supplies the control-and-behavior side of the story: intelligence as physically realized organization, behavior as transitions among activities, sensing and acting as core capacities, and internal control as the basis of outward competence. Heiserman does not replace that framework so much as deepen it by adding mechanisms

for experiential revision.

The difference, then, is not that Berkeley lacks embodiment. The difference is that Berkeley’s embodiment is primarily engineered and prescribed, whereas Heiserman’s becomes increasingly adaptive and experiential. Berkeley links logic and circuitry to organized machine behavior; Heiserman pushes the same broad architectural space toward creature-like adaptation. Their relation is therefore best understood as architectural continuity with developmental divergence.

4 A Contemporary Use-Case

Recent work suggests that this state-centered and behavior-organizing perspective still has technical relevance. Christopher A. Tucker’s 2026 arXiv paper, *A Behavioral State Vocabulary in Sony ERS-111 R-CODE*, argues from a corpus-level analysis of AIBO behavior diagrams that many superficially different routines are structured by a compact embodied grammar organized around initialization, sensing, iterative action, synchronization, and recovery [19]. Tucker further argues that this kind of state-based abstraction is useful as an intermediate representation for constructing new encapsulated behavior routines on constrained native robotic systems. That is significant here because it identifies a contemporary engineering setting in which explicit behavioral structuring remains a practical design resource.

This is important for the present argument because it shows a contemporary use-case in which behavioral state vocabularies are not treated as relics. They are treated as practically useful abstractions for embodied control. Tucker does not cite Berkeley or Heiserman as direct sources in the available abstract material, and no strong claim of influence is warranted here. But the structural resonance is striking. What Berkeley formalized as states, events, control, and organized machine activity, and what Heiserman developed into adaptive creature architectures, reappears as a useful representational strategy in modern constrained robotics.

The result is not proof, but it is more than a loose analogy. It supports the claim that the Berkeley–Heiserman line can be read as a still-open research program in which explicit behavior organization remains technically meaningful, especially where embodied systems must work under structural and computational constraints.

Berkeley should not be reduced to a symbolic logician who happened to mention machines. His work deserves to be read as an effort to show how intelligence can be built into physically organized systems that sense, remember, control, and act over time. His taxonomy of intelligent machines, his five-part architectural scheme, his extension of logic into states and events, and his robot examples together support a reading of Berkeley as a contributor to embodied machine intelligence in a control-oriented sense. What emerges from this reading is not a marginal curiosity, but an identifiable account of how intelligent conduct can be organized in machinery.

This conclusion should be stated carefully. Berkeley does not formulate adaptive, reinforcement-based, or homeostatic machine intelligence in the Heiserman manner. But that limitation clarifies his importance rather than diminishing it. Berkeley shows how formal logical structure can govern

temporally extended, physically embodied activity, while Heiserman shows how such organization can be pushed toward adaptive creature behavior. The value of the comparison is therefore not to flatten the two writers into one position, but to display a broader architecture whose fixed and adaptive variants can both be articulated.

The broader implication is that Berkeley and Heiserman together define more than a sequence of old publications. They define an underexplored engineering architecture: one in which logic, explicit state organization, embodied control, behavioral modularity, and adaptive response are treated as compatible ingredients of machine intelligence. Contemporary work on behavioral state vocabularies for constrained robotic systems lends support to taking that architecture seriously as a living research direction rather than as a closed curiosity [19]. The proper conclusion, then, is not that this approach has already been vindicated, but that it deserves renewed and systematic experimental attention as an alternative to the assumption that modern AI must choose between formal structure and embodied adaptive behavior.

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